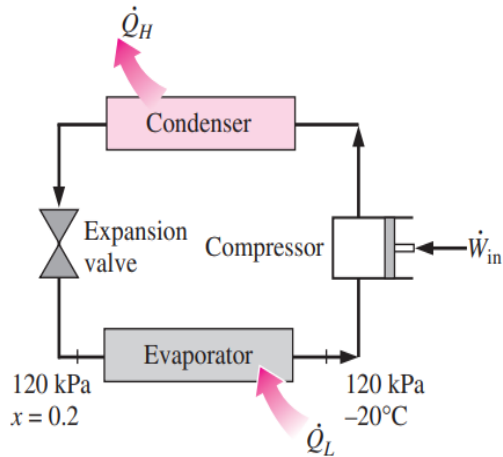
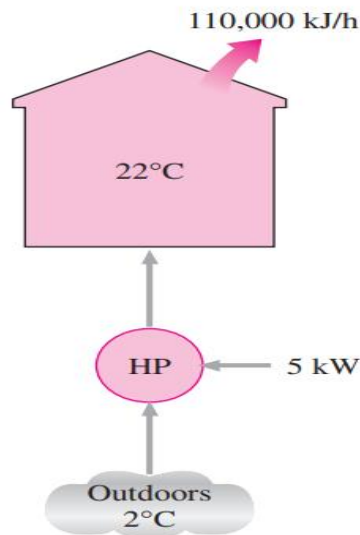


	BAHIR DAR INSTITUTE OF TECHNOLOGY (BiT) FACULTY OF MECHANICAL AND INDUSTRIAL ENGINEERING 2nd year Mechanical Engineering students Engineering Thermodynamics I Group Assignment (13%) Submission date: 16/09/2017 E.C	
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1. What is the purpose of introducing the second law of thermodynamics? Does the process occur only when it satisfies the second law of thermodynamics? Discuss this in comparison to the first law of thermodynamics.
2. Can we save all the heat rejected (Q_{out}) in a steam power plant? Discuss with diagram
3. Show that the ENTROPY (S) is state function(property).
4. Discuss the difference between Entropy and Energy.
5. Prove the second Carnot principle which states “The efficiencies of all reversible heat engines operating between the same two reservoirs are the same”.
6. A Carnot heat engine receives heat from a reservoir at 900°C at a rate of 800 kJ/min and rejects the waste heat to the ambient air at 27°C . The entire work output of the heat engine is used to drive a refrigerator that removes heat from the refrigerated space at -5°C and transfers it to the same ambient air at 27°C . Determine (a) the maximum rate of heat removal from the refrigerated space and (b) the total rate of heat rejection to the ambient air.
7. In choosing between an electrical heater and heat pump for heating the room, which one is preferable in terms of energy efficiency and environmental impact? State your reason clearly.
8. Why are engineers interested in reversible processes? Please explain.
9. Refrigerant-134a enters the evaporator coils placed at the back of the freezer section of a household refrigerator at 120 kPa with a quality of 20 percent and leaves at 120 kPa and -20°C . If the compressor consumes 450 W of power and the COP the refrigerator is 1.2, determine (a) the mass flow rate of the refrigerant and (b) the rate of heat rejected to the kitchen air.



10. A heat pump is used to maintain a house at 22°C by extracting heat from the outside air on a day when the outside air temperature is 2°C . The house is estimated to lose heat at a rate of $110,000 \text{ kJ/h}$, and the heat pump consumes 5 kW of electric power when running. Is this heat pump powerful enough to do the job?



11. An insulated piston–cylinder device contains 5 L of saturated liquid water at a constant pressure of 150 kPa . An electric resistance heater inside the cylinder is now turned on, and 2200 kJ of energy is transferred to the steam. Determine the entropy change of the water during this process.



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12. Helium gas is compressed from 90 kPa and 30°C to 450 kPa in a reversible, adiabatic process. Determine the final temperature and the work done, assuming the process takes place (a) in a piston–cylinder device and (b) in a steady-flow compressor. (Given constant for Helium: $C_v = 3.1156$ kJ/kg.K, $C_p = 5.1926$ kJ/kg.K, and $k = 1.667$)
13. Steam enters an adiabatic turbine at 8 MPa and 500°C with a mass flow rate of 3 kg/s and leaves at 30 kPa. The isentropic efficiency of the turbine is 0.90. Neglecting the kinetic energy change of the steam, determine (a) the temperature at the turbine exit and (b) the power output of the turbine.

